

Hungjui Yu

Cooperative Institute for Research in the Atmosphere (CIRA) | Department of Atmospheric Science
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+ CURRENT POSITION

Research Scientist

Cooperative Institute for Research in the Atmosphere (CIRA), Colorado State University (CSU)
Fort Collins, CO, USA

+ RESEARCH TOPICS

- ⦿ 3D Cloud Structure and Variability in the Current and Future Climate
- ⦿ Mesoscale Convective Process and the Environments using Machine Learning
- ⦿ Application of Remote Sensing to Satellite Missions, Aviation and Severe Weather Hazards
- ⦿ Global Spatial and Temporal Variability of Mesoscale Convective Systems & Organized Convection
- ⦿ Observational Field Campaigns, and Radar/Radiosonde Operations, Data Quality Control and Analysis

+ EDUCATION

- ⦿ 2011 – 2018 **Ph.D. in Atmospheric Sciences** > Department of Atmospheric Sciences, National Taiwan University
Dissertation: “Quasi-2-day Convective Disturbances in the Equatorial Indian Ocean: DYNAMO Observation”
Advisor: Dr. Hung-Chi Kuo and Dr. Richard H. Johnson
- ⦿ 2012 – 2014 **Visiting Student** > Department of Atmospheric Science, Colorado State University
Graduate Student Study Abroad Program, Ministry of Science and Technology, Taiwan
Advisor: Dr. Richard H. Johnson
- ⦿ 2005 – 2009 **B.S. in Atmospheric Sciences** > Department of Atmospheric Sciences, National Taiwan University
Dean’s Award

+ PROFESSIONAL EXPERIENCE

- ⦿ 2025 – present **Research Scientist/Scholar I: Cloud Properties & Cloud-Free Line of Sight Specialist**
Cooperative Institute for Research in the Atmosphere (CIRA), Colorado State University (CSU)
 - OVERCAST Project
Major developer of the Deterministic Cloud-Free Line-of-Sight (DCFLOS) application with the CIRA 3D cloud dataset.

- INCUS Project
Major developer of the auxiliary lightning (AUX-LIGHT) product and co-lead of level 4 satellite ephemeris (4-ORBIT) product.
- CAIG Project
Co-I and major contributor to Application 2.
- NOAA MAPP Project
Co-I and Objective 1 lead.

◎ 2023 – 2025

Postdoctoral Fellow (extended offer of Research Scientist 1)

Cooperative Institute for Research in the Atmosphere (CIRA), Colorado State University (CSU)

- RAM-HORNS/OVERCAST Project
Major developer for the DCFLOS application with CIRA 3D cloud dataset under RAM-HORNS/OVERCAST projects.

◎ 2020 – 2023

Postdoctoral Fellow

Department of Atmospheric Science, Colorado State University

- Field Campaign: Prediction of Rainfall Extremes Campaign in the Pacific (PRECIP) – pre-experiment 2021
Participation in Radiosonde operation, radar strategy decision-making, personnel training, and instrument development for the campaign and radiosonde network
- Package development: Cloud System Classification
Major developer of the TRMM-heritage Storm Mode classification Python package for multiple observation and numerical model datasets.

◎ 2018 – 2020

Postdoctoral Fellow

Department of Atmospheric Sciences, National Taiwan University

- Field Campaign: Taipei Severe Storm Experiment (TASSE)
Major field campaign coordinator for decision-making, weather analysis, personnel training, and instrument development for the radiosonde network.

◎ 2011 – 2018

Student Research Assistant

Research Center of Climate Change and Sustainable Development, National Taiwan University

+ GRANTS AND FELLOWSHIP

◎ 2023 – present

U.S. Office of Naval Research (ONR) RAM-HORNS (Research Advances in Meteorology-Honing Operational Readiness for National Security) and Optical Variability Evaluation of Regional Cloud Asymmetries in Space and Time (OVERCAST)
As the lead developer of the Deterministic Cloud-Free Line-of-Sight (DCFLOS) algorithm and tool in Task 5.
PI: Dr. Steven D. Miller (CSU)

- ⦿ 2024 – present **National Science Foundation (NSF) Collaborations in Artificial Intelligence and Geosciences (CAIG) Program (Award #: 2425923)**
 “Toward a Deeper Understanding of Cloud Processes and Future Storm Modes using AI”
 As co-I and major contributor to the Application 2.
 PI: Dr. Imme Ebert-Uphoff (CSU)
- ⦿ 2024 – present **INvestigation of Convective Updrafts (INCUS) Mission**
 As major algorithm developer for AUX-LIGHT product and co-lead of 4-ORBIT product.
 PI: Dr. Susan van den Heever (CSU)
 DPI: Dr. Kristen Rasmussen (CSU)
- ⦿ 2024 – present **NOAA Climate Program Office (CPO), Modeling, Analysis, Predictions, and Projections (MAPP) Program**
 “Storm Mode Classification as a Process-Oriented Tool to Diagnose Precipitation Biases in Climate Models”
 As co-I and the Objective 1 lead.
 PI: Dr. Kristen Rasmussen (CSU)
- ⦿ 2020 – 2021 **Postdoctoral Research Abroad Program, Ministry of Science and Technology, Taiwan**
 “Characteristics and Mechanisms for Mesoscale Convective Systems and Rainfall Extremes in the Tropical Ocean and Land”
- ⦿ 2013 – 2014 **Graduate Student Study Abroad Program, Ministry of Science and Technology, Taiwan**
 “Characteristics and Variability of Atmospheric Stable Layers during DYNAMO-AMIE 2011”

+ PUBLICATIONS

1. Tianqi Zuo, Alison D. Nugent, Angela K. Rowe, Kristen L. Rasmussen, **H. Yu**, 2026: Climatology of Taiwan's Heavy Rainfall Events and Associated Storm Types. (*submitted to Monthly Weather Review*)
2. Dongwei Chen, Emily J. King, **H. Yu**, 2026: Taylor Diagram and Wasserstein Distance for Model Evaluation. (*submitted to Sampling Theory, Signal Processing, and Data Analysis*)
3. **Yu, H.**, Ver Hoef, L., Rasmussen, K. L., Ebert-Uphoff, I. (2026). Exploring the Ability of Machine Learning to Infer Subgrid-scale Convective Cloud Fraction from Coarse-Resolution Environmental Conditions. *Artificial Intelligence for the Earth Systems*. <https://doi.org/10.1175/AIES-D-25-0085.1>
4. **Yu, H.**, Rasmussen, K. L., Dolan, B. (2025). Current and Future Convective Storm Modes over CONUS from GPM Observations and Convection-Permitting Regional Climate Model Simulations. (*reformatting for submission*)
5. H.-C. Kuo, T.-S. Yo, **H. Yu**, S.-H. Su, C.-H. Liu, P.-H. Lin (2025). Data Quality Control and Calibration for Mini-Radiosonde System “Storm Tracker” in Taiwan. *Journal of*

the Meteorological Society of Japan Ser II (気象集誌. 第2輯), 103(5), 573–593. doi: 10.2151/jmsj.2025-029.

6. **Yu, H.**, Rasmussen, K. L., Kuo, H.-C. (2021). Quasi-2-day and diurnal cloud variation timescales over convectively active regions. *Journal of Geophysical Research: Atmospheres*, 126, e2021JD035426. <https://doi.org/10.1029/2021JD035426>
7. Tsujino, S., H.-C. Kuo, **H. Yu**, B.-F. Chen, and K. Tsuboki (2021). Effects of mid-level moisture and environmental flow on the development of afternoon thunderstorms in Taipei. *Terr. Atmos. Ocean. Sci.*, 32, 497-518, doi: 10.3319/TAO.2021.11.17.01.
8. Hwang, W. C., Lin, P. H., & **Yu, H.** (2020). The development of the “Storm Tracker” and its applications for atmospheric high-resolution upper-air observations. *Atmospheric Measurement Techniques*, 13(10), 5395-5406. <https://doi.org/10.5194/amt-13-5395-2020>
9. **Yu, H.**, Johnson, R. H., Ciesielski, P. E., & Kuo, H. C. (2018). Observation of quasi-2-day convective disturbances in the equatorial Indian Ocean during DYNAMO. *Journal of the Atmospheric Sciences*, 75(9), 2867-2888. <https://doi.org/10.1175/JAS-D-17-0351.1>
10. **Yu, H.**, Ciesielski, P. E., Wang, J., Kuo, H. C., Vömel, H., & Dirksen, R. (2015). Evaluation of humidity correction methods for Vaisala RS92 tropical sounding data. *Journal of Atmospheric and Oceanic Technology*, 32(3), 397-411. <https://doi.org/10.1175/JTECH-D-14-00166.1>
11. Ciesielski, P. E., **Yu, H.**, Johnson, R. H., Yoneyama, K., Katsumata, M., Long, C. N., ... & Van Hove, T. (2014). Quality-controlled upper-air sounding dataset for DYNAMO/CINDY/AMIE: Development and corrections. *Journal of Atmospheric and Oceanic Technology*, 31(4), 741-764. <https://doi.org/10.1175/JTECH-D-13-00165.1>

+ INVITED PRESENTATIONS

- ◎ 2026 **IPC15 Pre-conference Asia Virtual Workshop**
Bridging Weather, Climate, and AI/ML for Severe Storms
- ◎ 2023 **Central Weather Bureau (now Central Weather Administration, CWA) in Taiwan**
Clouds and Storms on Bridging Weather and Climate
- ◎ 2022 **Climate Hotspots in Action (CHiA) Forum Webinar**
Quasi-Two-Day and Diurnal Cloud Variation Timescales over Convectively Active Regions
- ◎ 2020 **Department of Atmospheric Sciences, National Central University (NCU)**
Department of Atmospheric Sciences, Chinese Culture University (PCCU)
Quality-Controlled High-Resolution Upper-Air Sounding Dataset for TASSE: Development and Corrections of the “Storm Tracker” Observations

- ◎ 2019
Department of Atmospheric Sciences, Chinese Culture University (PCCU)
 Taipei Severe Storm Experiment (TASSE): Upper-air Radiosonde Observations and the Development of the “Storm Tracker”
- ◎ 2018
Department of Earth Sciences, National Taiwan Normal University (NTNU)
 Quasi-2-day Convective Disturbances in the Equatorial Indian Ocean: DYNAMO Observation

+ CONFERENCE PRESENTATIONS

- ◎ 2026
The 17th International Conference on Mesoscale Convective Systems (ICMCS-XVII)
 1. Using machine learning to downscale coarse-resolution environmental variables for understanding the spatial frequency of convective storms
 2. Leveraging Machine Learning to Study Storms for Scientific Discovery (presented by K. L. Rasmussen)
- ◎ 2023
The 15th International Conference on Mesoscale Convective Systems (ICMCS-XV)
 Current & Future Convective Storm Modes over CONUS from GPM Observations and Convection-permitting Regional Climate Model Simulations
- ◎ 2022
American Meteorological Society (AMS) 102nd Annual Meeting
 1. Quasi-Two-Day and Diurnal Cloud Variation Timescales over Convectively Active Regions
 2. Upper-Air Radiosonde Observations and Data Corrections of the Storm Tracker during PRECIP 2021
- ◎ 2020
2020 Conference on Weather Analysis and Forecasting (Central Weather Bureau, Taiwan)
 Quality-Controlled High-Resolution Upper-Air Sounding Dataset for TASSE: Development and Corrections of the Storm Tracker Observations
- ◎ 2019
Asia Oceania Geosciences Society (AOGS) 16th Annual Meeting
 Quasi-2-day Convective Disturbances Over the Equatorial Indian Ocean and Western Pacific

2019 Taipei Severe Weather and Extreme Precipitation (SWEP) Workshop
 Upper-air Radiosonde and “Storm Tracker” Observations in TASSE 2018

13th International Conference on Mesoscale Convective Systems and High-Impact Weather in East Asia (ICMCS-XIII)
 Characteristics of Quasi-2-Day Convective Disturbances over the Tropical Ocean
- ◎ 2017
12th International Conference on Mesoscale Convective Systems and High-Impact Weather in East Asia (ICMCS-XII)
 An Observational Study on Quasi-2-day Convective Disturbances in the Equatorial Indian Ocean during DYNAMO/AMIE/CINDY 2011
- ◎ 2015
2015 American Geophysical Union (AGU) Fall Meeting
 Two-day Convective Disturbances in the Equatorial Indian Ocean

2015 International Workshop on Typhoon and Flood–APEC Experience Sharing on Hazardous Weather Events and Risk Management
Two-day Disturbances over the Equatorial Indian Ocean during DYNAMO-AMIE-CINDY 2011

◎ 2014

Asia Oceania Geosciences Society (AOGS) 11th Annual Meeting
Characteristics and Variability of the Melting Stable Layer during DYNAMO-AMIE-CINDY 2011

17th Symposium on Meteorological Observations and Instrumentation / 21st Conference on Applied Climatology
Evaluation of GRUAN and DigiCORA Humidity Corrections to Vaisala RS92 Sounding Data during DYNAMO

2014 National Conference of Graduate Students in Atmospheric Sciences
Characteristics and Variability of the Melting Stable Layer during DYNAMO-AMIE-CINDY 2011

◎ 2013

2013 American Geophysical Union (AGU) Fall Meeting
Characteristics and Variability of the Melting Stable Layer during DYNAMO-AMIE-CINDY 2011

◎ 2012

2012 American Geophysical Union (AGU) Fall Meeting
Validation of Sonde Moisture Corrections using GPS and MWR Precipitable Water Retrievals during DYNAMO-CINDY 2011-AMIE

+ CODE PACKAGES DEVELOPMENT

◎ 2026

DCFLOS_Toolbox: v0.5.3
Python package designed to estimate the DCFLOS using a 3D cloud dataset generated at CIRA. Task 5 for the OVERCAST projects. Version 0.5.3 has been delivered to ONR.

◎ 2022

Cloud System Classification: v1.0
Fundamental tool for classifying cloud systems for CAIG project.
Yu, H. (2022). yuhungjui/Cloud_System_Classification: v1.0 (v1.0). Zenodo.
<https://doi.org/10.5281/zenodo.6491940>

+ PROFESSIONAL SERVICE AND OUTREACH

◎ Services

Associate Editor (2026 – present)
Artificial Intelligence for the Earth Systems (AIES) (ISSN: 2769-7525)

Journal Peer Reviews
Artificial Intelligence for the Earth Systems (AIES) (ISSN: 2769-7525)
Asia-Pacific Journal of Atmospheric Sciences (ISSN: 1976-7951)
Bulletin of the American Meteorological Society (BAMS) (ISSN: 1520-0477)
Climate Dynamics (ISSN: 0930-7575)

Journal of Atmospheric and Oceanic Technology (JAOT) (ISSN: 1520-0426)
 Journal of Geophysical Research: Atmospheres (JGR: Atmospheres) (ISSN: 2169-8996)
 Monthly Weather Review (MWR) (ISSN: 1520-0493)
 Quarterly Journal of the Royal Meteorological Society (QJRMS) (ISSN: 1477-870X)

⦿ Conferences and Workshops

2025 Taiwan-US Workshop on AI/ML for Satellite Data, Severe Weather, and Tropical Cyclones at CIRA

CSU side of the workshop was co-organized by Dr. Hungjui Yu (ATS/CIRA), Dr. Imme Ebert-Uphoff (CIRA) and Dr. Michael Bell (ATS).

⦿ Student Mentoring

Mesoscale & Climate Research Group (Rasmussen Research Group) at Department of Atmospheric Science at CSU

Scientific discussion and co-advising graduate students of Dr. Kristen Rasmussen at Department of Atmospheric Science at CSU.

+ HONORS AND AWARDS

⦿ 2015

Outstanding Student Poster Award in International Conference

2015 American Geophysical Union (AGU) Fall Meeting

1st Place of IWTF Student Poster Competition Award

2015 International Workshop on Typhoon and Flood (IWTF), Taipei, Taiwan

⦿ 2014

Distinction in Student Poster Session

2014 National Conference of Graduate Students in Atmospheric Sciences, Central Weather Bureau (CWB), Taiwan

⦿ 2009

Dean's Award

College of Science, National Taiwan University (NTU)

+ FIELD CAMPAIGNS

⦿ 2021

Prediction of Rainfall Extremes Campaign in the Pacific (PRECIP) – pre-experiment 2021

Radiosonde operation, radar strategy decision-making, personnel training, and instrument development for the campaign and radiosonde network

⦿ 2016 – 2020

TAIpei Severe Storm Experiment (TASSE)

Organization, decision-making, weather analysis, personnel training, and instrument development for the campaign and radiosonde network

⦿ 2011 – 2012

Dynamics of Madden-Julian Oscillation (DYNAMO)

Field operator conducting upper-air radiosonde observation at Malé, Maldives